

V-Dem Data Lab

Exploring V-Dem and V-Party in R

[Your Names]

2026-10-09

Getting started

This file is a toolbox for your individual exploration. Run the examples, then change the countries, years or indicators to follow your own interests. The aim is to understand a measure, discover a pattern and develop a research question. You do not need to work through every example.

For our ~45-minute session, spend about 35 minutes exploring and keep the final 10 minutes for discussion. Choose either the V-Dem or the V-Party example.

In RStudio, run a code chunk using its green arrow. In Positron, use the >Run Cell button above each code cell. **Render** produces a document from the whole file. Write your observations in the spaces below.

Preparation

Install packages once

Run this chunk manually in the beginning of the session. It is excluded from rendering (`eval: false`). Installation requires an internet connection and may take a few moments. See the official package repository for installation guidance.

```
install.packages(c("tidyverse", "remotes"))
remotes::install_github("vdeminstitute/vdemdata")
```

Load packages and datasets

```
if (!require(pacman)) install.packages("pacman")
p_load(tidyverse, vdemdata)

vdem_data <- vdemdata::vdem
vparty_data <- vdemdata::vparty
codebook <- vdemdata::codebook #vdem codebook only, vparty codebook available:
```

```
# Record the installed package version.
packageVersion("vdemdata")
```

```
[1] '16.0'
```

V-Dem here uses **country-year** observations. V-Party uses **country-party-date** observations: do not assume that each party is observed every year. Check the coverage of your chosen sample before comparing it. The datasets have different time coverage and release schedules.

1. Inspect the data and find a measure

Basic tools

```
# Number of rows and columns
dim(vdem_data)
```

```
[1] 28092  4618
```

```
dim(vparty_data)
```

```
[1] 11898   384
```

```
# A small preview, rather than printing the entire dataset
vdem_data |> select(country_name, year, v2x_polyarchy) |> head()
```

	country_name	year	v2x_polyarchy
1	Mexico	1789	0.027
2	Mexico	1790	0.027
3	Mexico	1791	0.027
4	Mexico	1792	0.027
5	Mexico	1793	0.027
6	Mexico	1794	0.027

```
vparty_data |> select(country_name, year, v2paenname, v2paid) |> head()
```

	country_name	year	v2paenname	v2paid
1	Mexico	1991	Party of the Democratic Revolution	216
2	Mexico	1994	Party of the Democratic Revolution	216
3	Mexico	1997	Party of the Democratic Revolution	216

4	Mexico 2000	Party of the Democratic Revolution	216
5	Mexico 2003	Party of the Democratic Revolution	216
6	Mexico 2006	Party of the Democratic Revolution	216

```
# Available years
range(vdem_data$year, na.rm = TRUE)
```

```
[1] 1789 2025
```

```
range(vparty_data$year, na.rm = TRUE)
```

```
[1] 1900 2019
```

For an interactive spreadsheet-style view, use `view()`. These commands are optional and excluded from rendering.

```
view(codebook)
view(vdem_data)
view(vparty_data)
```

Search the V-Dem codebook

```
# Search indicator names by keyword; replace the search term.
democracy_variables <- vdemdata::find_var("democracy")
democracy_variables |> select(tag, name) |> head(10)
```

	tag	name
1	e_boix_regime	Democracy (BMR)
2	e_chga_demo	Democracy
3	e_democ	Institutionalized democracy
4	e_uds	Unified democracy score posterior
5	e_v2x_delibdem	Deliberative democracy index ordinal
6	e_v2x_egaldem	Egalitarian democracy index ordinal
7	e_v2x_libdem	Liberal democracy index ordinal
8	e_v2x_partipdem	Participatory democracy index ordinal
9	e_v2x_polyarchy	Electoral democracy index ordinal
10	v2cademmob	Mobilization for democracy

```
# Ask about a specific variable using its name in quotation marks.  
vdemdata::var_info("v2x_polyarchy")
```

```
$name  
[1] "Electoral democracy index"  
  
$vartype  
[1] "D"  
  
$tag  
[1] "v2x_polyarchy"  
  
$projectmanager  
[1] "Jan Teorell"  
  
$question  
[1] "To what extent is the ideal of electoral democracy in its fullest sense  
achieved?"  
  
$clarification  
[1] "The electoral principle of democracy seeks to embody the core value of  
making rulers responsive to citizens, achieved through electoral competition  
for the electorate's approval under circumstances when suffrage is extensive;  
political and civil society organizations can operate freely; elections are  
clean and not marred by fraud or systematic irregularities; and elections affect  
the composition of the chief executive of the country. In between elections,  
there is freedom of expression and an independent media capable of presenting  
alternative views on matters of political relevance. In the V-Dem conceptual  
scheme, electoral democracy is understood as an essential element of any other  
conception of representative democracy --- liberal, participatory, deliberative,  
egalitarian, or some other."  
  
$responses  
[1] NA  
  
$scale  
[1] "Interval, from low to high (0-1)."  
  
$notes  
[1] NA  
  
$crosscoder_aggregation
```

```

[1] NA

$cy_aggregation
[1] "Day-weighted mean"

$datarelease
[1] "1-16. Release 1-5 used a different, preliminary aggregation formula."

$years
[1] "1789-2025"

$convergence
[1] NA

```

`find_var()` returns a table of codebook entries, not just variable names. `var_info()` expects a variable name, not a dataset. These helpers are documented for the V-Dem codebook. For V-Party, consult its separate codebook on the dataset page.

You can search column names in either dataset. This finds names, not definitions:

```
names(vdem_data)[str_detect(names(vdem_data), "^v2x_")] |> head(15)
```

```

[1] "v2x_polyarchy"      "v2x_polyarchy_codelow" "v2x_polyarchy_codehigh"
[4] "v2x_polyarchy_sd"   "v2x_libdem"            "v2x_libdem_codelow"
[7] "v2x_libdem_codehigh" "v2x_libdem_sd"         "v2x_partipdem"
[10] "v2x_partipdem_codelow" "v2x_partipdem_codehigh" "v2x_partipdem_sd"
[13] "v2x_delibdem"       "v2x_delibdem_codelow"  "v2x_delibdem_codehigh"

```

```
names(vparty_data)[str_detect(names(vparty_data), "vote|seat|corrupt")]
```

```

[1] "v2paseatshare"      "v2panumbseat"          "v2patotalseat"
[4] "v2pavote"           "ep_corrupt_salience"

```

Your notes: Which variable interests you? What does it measure? What do high and low values mean? What aspect of your interest does it leave out?

Write here.

2A. V-Dem: Compare countries over time

Select countries, years and columns

```
# Change these settings to explore your own cases.
chosen_countries <- c("Germany", "Hungary", "Poland")
first_year <- 1990
last_year <- 2024

democracy_sample <- vdem_data |>
  filter(country_name %in% chosen_countries,
         year >= first_year, year <= last_year) |>
  select(country_name, year, v2x_polyarchy, v2x_libdem)

head(democracy_sample)
```

	country_name	year	v2x_polyarchy	v2x_libdem
1	Poland	1990	0.740	0.660
2	Poland	1991	0.866	0.783
3	Poland	1992	0.888	0.804
4	Poland	1993	0.887	0.808
5	Poland	1994	0.890	0.811
6	Poland	1995	0.889	0.810

```
# Check how many observations and missing values each country has.
democracy_sample |>
  group_by(country_name) |>
  summarise(observations = n(),
            missing_electoral = sum(is.na(v2x_polyarchy)))
```

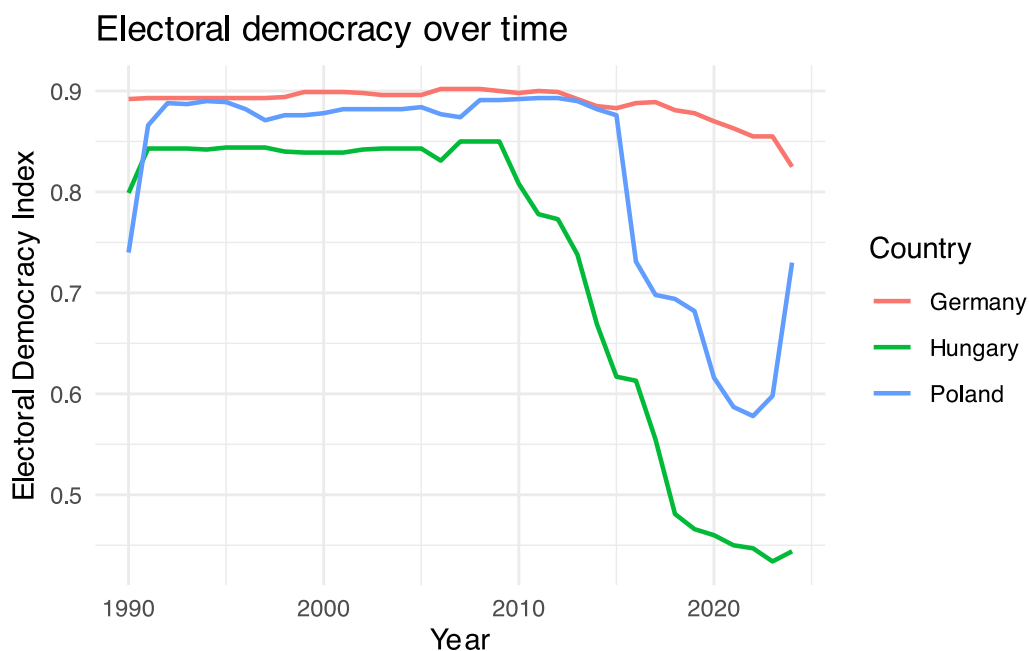
```
# A tibble: 3 × 3
  country_name observations missing_electoral
  <chr>           <int>           <int>
1 Germany         35             0
2 Hungary         35             0
3 Poland          35             0
```

`filter()` selects rows; `select()` selects columns. `%in%` matches any of the countries in your list. `|>` passes the result to the next step. `NA` means missing data, not zero.

Draw a line graph

```
democracy_plot <- ggplot(democracy_sample,
  aes(x = year, y = v2x_polyarchy,
      colour = country_name)) +
  geom_line(linewidth = 0.8) +
  labs(title = "Electoral democracy over time",
    x = "Year", y = "Electoral Democracy Index",
    colour = "Country") +
  theme_minimal()

democracy_plot
```



```
var_info("v2x_polyarchy")
```

```
$name
[1] "Electoral democracy index"

$vartype
[1] "D"

$tag
[1] "v2x_polyarchy"
```

\$projectmanager

[1] "Jan Teorell"

\$question

[1] "To what extent is the ideal of electoral democracy in its fullest sense achieved?"

\$clarification

[1] "The electoral principle of democracy seeks to embody the core value of making rulers responsive to citizens, achieved through electoral competition for the electorate's approval under circumstances when suffrage is extensive; political and civil society organizations can operate freely; elections are clean and not marred by fraud or systematic irregularities; and elections affect the composition of the chief executive of the country. In between elections, there is freedom of expression and an independent media capable of presenting alternative views on matters of political relevance. In the V-Dem conceptual scheme, electoral democracy is understood as an essential element of any other conception of representative democracy --- liberal, participatory, deliberative, egalitarian, or some other."

\$responses

[1] NA

\$scale

[1] "Interval, from low to high (0-1)."

\$notes

[1] NA

\$crosscoder_aggregation

[1] NA

\$cy_aggregation

[1] "Day-weighted mean"

\$datarelease

[1] "1-16. Release 1-5 used a different, preliminary aggregation formula."

\$years

[1] "1789-2025"

\$convergence

[1] NA

```
var_info("v2x_libdem")
```

```
$name
```

```
[1] "Liberal democracy index"
```

```
$vartype
```

```
[1] "D"
```

```
$tag
```

```
[1] "v2x_libdem"
```

```
$projectmanager
```

```
[1] "Jan Teorell"
```

```
$question
```

```
[1] "To what extent is the ideal of liberal democracy achieved?"
```

```
$clarification
```

```
[1] "The liberal principle of democracy emphasizes the importance of protecting individual and minority rights against the tyranny of the state and the tyranny of the majority. The liberal model takes a \"negative\" view of political power insofar as it judges the quality of democracy by the limits placed on government. This is achieved by constitutionally protected civil liberties, strong rule of law, an independent judiciary, and effective checks and balances that, together, limit the exercise of executive power. To make this a measure of liberal democracy, the index also takes the level of electoral democracy into account."
```

```
$responses
```

```
[1] NA
```

```
$scale
```

```
[1] "Interval, from low to high (0-1)."
```

```
$notes
```

```
[1] NA
```

```
$crosscoder_aggregation
```

```
[1] NA
```

```
$cy_aggregation
```

```
[1] "Day-weighted mean"
```

```

$datarelease
[1] "1-16. Release 1, 2, and 3 used a different, preliminary aggregation formula."

$years
[1] "1789-2025"

$convergence
[1] NA

```

Try it: Replace `v2x_polyarchy` in the plot with `v2x_libdem`, and adjust the title and axis label. Consult `var_info("v2x_libdem")` first. These indices capture different concepts; a difference between their values is not itself a measure of institutional change.

Your notes: Describe one pattern. Does it match your expectations? Separate the observation from your possible explanation.

Write here.

2B. V-Party: Compare parties at one election date

Party names are stored in `v2paennname` (English name), and party identifiers in `v2paid`. The example uses vote shares, `v2pavote`; consult the V-Party codebook for its definition, scale and treatment of electoral alliances.

```

chosen_country <- "Germany"

# Inspect available dates before choosing one.
vparty_data |>
  filter(country_name == chosen_country) |>
  distinct(year, historical_date) |>
  arrange(desc(year), historical_date) |>
  head(10)

```

	year	historical_date
1	2017	2017-09-24
2	2013	2013-09-22
3	2009	2009-09-27
4	2005	2005-09-18
5	2002	2002-09-22
6	1998	1998-09-27
7	1994	1994-10-16
8	1990	1990-12-02
9	1987	1987-01-25
10	1983	1983-03-06

```
# Example: German parties observed in 2017.
party_sample <- vparty_data |>
  filter(country_name == chosen_country, year == 2017) |>
  select(country_name, year, historical_date,
         v2paid, v2paenname, v2pavote)

party_sample
```

	country_name	year	historical_date	v2paid
1	Germany	2017	2017-09-24	211
2	Germany	2017	2017-09-24	383
3	Germany	2017	2017-09-24	573
4	Germany	2017	2017-09-24	1375
5	Germany	2017	2017-09-24	1545
6	Germany	2017	2017-09-24	1731
7	Germany	2017	2017-09-24	1816
8	Germany	2017	2017-09-24	1976

		v2paenname	v2pavote
1	Christian Democratic Union / Christian Social Union		NA
2	Social Democratic Party of Germany		20.5
3	Free Democratic Party		10.7
4	Christian Democratic Union		28.2
5	The Left		9.2
6	Christian Social Union		6.5
7	Alliance 90 / Greens		8.9
8	Alternative for Germany		12.6

Check that your sample refers to one election date before making the chart. If your chosen year contains several dates, add a filter for `historical_date`. Missing parties or missing vote shares should not be interpreted as zero votes.

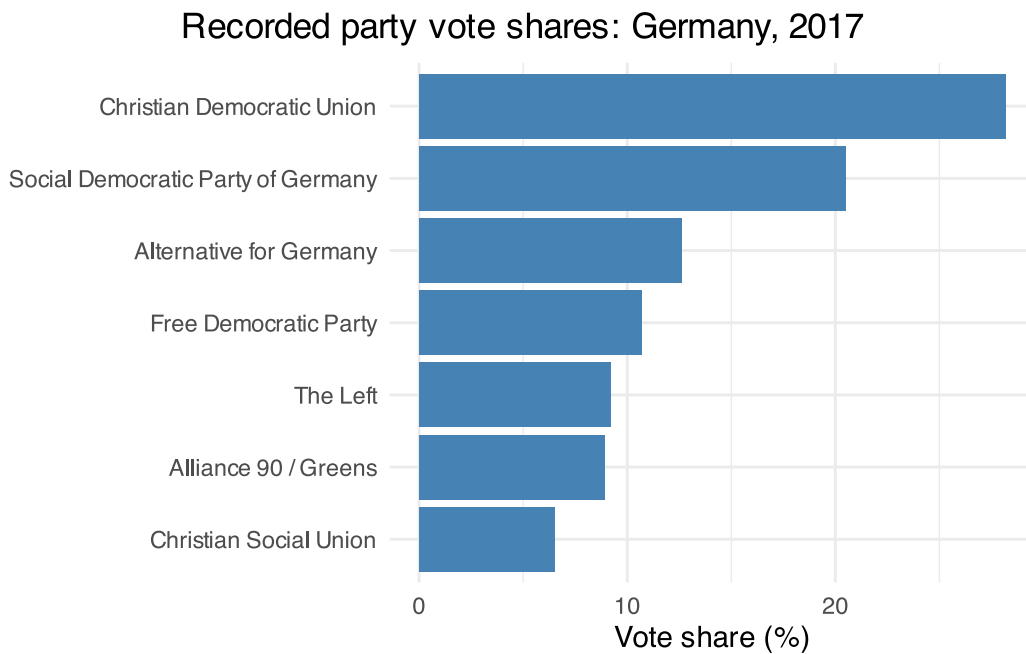
```
party_sample |> summarise(missing_vote_shares = sum(is.na(v2pavote)))
```

```
missing_vote_shares
1                    1
```

```
party_plot <- ggplot(party_sample |> filter(!is.na(v2pavote)),
  aes(x = reorder(v2paenname, v2pavote), y = v2pavote)) +
  geom_col(fill = "steelblue") +
  coord_flip() +
  labs(title = "Recorded party vote shares: Germany, 2017",
```

```
x = NULL, y = "Vote share (%)") +
theme_minimal() +
theme(
  plot.title.position = "plot",
  plot.title = element_text(hjust = 0.5)
)
```

party_plot



Try it: Choose another country and an available election year. Update the chart title. Alternatively, find a party characteristic in the V-Party codebook, locate its column and explore it. Use party identifiers when tracking parties; names alone may not reliably identify them over time.

Your notes: What does your comparison show? What would you need to know about the election or party coverage to interpret it?

Write here.

3. Develop a research question

Based on your exploration, write down:

1. One research question.
2. The countries, parties and period you would investigate.
3. One additional variable or data source you would need.
4. Something your current graph cannot establish.

A descriptive pattern can inspire an explanation, but does not establish causality. Where uncertainty intervals are provided, consider them before interpreting small differences.

Write here.

Optional tools: Save your results

Run manually if you want to save your V-Dem sample and plot in the folder containing this file. These commands are excluded from rendering.

```
write_csv(democracy_sample, "my-vdem-sample.csv")

ggsave("my-democracy-plot.png", plot = democracy_plot,
       width = 8, height = 5, dpi = 300)

ggsave("my-party-plot.png", plot = party_plot,
       width = 8, height = 5, dpi = 300)
```

Sources and reproducibility

- vdemdata package documentation
- V-Dem data and codebook
- V-Party data and codebook

Record the package and dataset versions you use and cite the datasets in research. Values may be revised between releases: use one release throughout your comparison. HTML is the default output; select Typst when you want a PDF.